

# ASHISH SURESH

## Embedded C++ Engineer

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GitHub: [github.com/ashu304-ops/project1](https://github.com/ashu304-ops/project1)

## Profile

Mechanical Engineer with 3 years of professional engineering experience transitioning into Embedded C++ development. Strong C++ foundation with hands-on project work in embedded C++, STM32 concepts, FreeRTOS, and Raspberry Pi software simulation. Seeking an entry-level/junior Embedded C++ or Firmware Engineer opportunity.

## Technical Skills

- **C++:** Modern C++, OOP, STL, Templates, RAII, Smart Pointers, C++17, FreeRTOS, STM32, Pointers, UART, SPI, I2C, Memory Management
- **Embedded & Protocols:** C, STM32 Architecture, GPIO, ADC, PWM, Timers, Interrupts, UART, SPI, I2C
- **RTOS & OS:** FreeRTOS (Tasks, Scheduling, Queues, Mutex, Semaphores, Software Timers), Linux Fundamentals, IPC
- **Development Tools:** CMake, GCC/GDB, Git, GitHub, Raspberry Pi Software Simulation
- **Concepts & Architecture:** SOLID Principles, Design Patterns, Debugging, Testing, Embedded Linux Integration

## Embedded Engineering Projects

### Embedded Vehicle Suspension & Safety Controller

[GitHub Repository](#)

*C++ — STM32 — FreeRTOS — Raspberry Pi Simulation*

- Developed an Embedded C++ vehicle control system with simulated hardware sensors, actuators, and safety mechanisms.
- Applied Object-Oriented Design, RAII, and STL to design modular C++ firmware components for real-time operations.
- Implemented FreeRTOS tasks, message queues, and synchronization primitives for sensor acquisition and safety controls.
- Configured virtual peripherals for STM32 concepts including GPIO, ADC, PWM, timers, interrupts, UART, SPI, and I2C.
- Built a Raspberry Pi software simulation environment to emulate hardware interactions, sensor failures, and thermal faults.
- Structured system architecture to prepare for future Embedded Linux integration and real-time visual monitoring.

## Professional Experience

### Dipesh Engineering Pvt. Ltd. – Design Engineer

**Mumbai, India**

*Professional Engineering (3 Years)*

- Designed complex engineering systems according to thermal, mechanical, and manufacturing specifications.
- Executed structured engineering analysis, parametric calculations, and technical optimization for clients.
- Collaborated with cross-functional manufacturing and engineering teams to resolve technical design edge cases.

## Education

### Bachelor of Engineering – Mechanical Engineering

**Mumbai, India**

Thakur College of Engineering and Technology